Type MS · Expressions for file size and transfer time

Answer 1 of 9 · Question 2 part d.ii · 4 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part d.ii in this mark scheme extract

2(a)(i)	Award up to **two** marks for: • 0100 (1) 1110 (1)	Accept any number of leading zeros.	2
2(a)(ii)	**The only correct answer is B** *A is not correct because standard ASCII uses 7 bits, 64 characters would need only 6 bits* *C is not correct because standard ASCII uses 7 bits, 256 characters would need 8 bits* *D is not correct because standard ASCII uses 7 bits, 512 characters would need 9 bits*		1
2(a)(iii)	Award up to **two** marks for a linked explanation such as: All of the major languages/symbols/characters can be represented by Unicode (1) because it uses a minimum of 16 bits/more bits/32 bits/65536 characters (1) Unicode can represent all/more characters/any language (1) whereas ASCII can only represent English/Latin/128 characters/doesn't have enough characters (1) Unicode can represent all characters (1) because it uses 16 bits/2 bytes /more bits instead of 8 bits/1 byte (1)	Accept the reverse argument Allow examples of non-latin characters	2

Question Number	Answer	Additional Guidance	Mark
2(b)	1010 1011 Award up to **two** marks for: • MSB = 1 (1) • Rest of pattern correct 010 1011 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(c)	Award up to **two** marks for: • 0100 (1) 1010 (1)		2

Type MS · Expressions for file size and transfer time

Answer 1 of 9 · Question 2 part d.ii · 4 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part d.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2(d)(i)</td><td>Award up to two marks for:

• 3 bits used for all patterns (1)/
• No pattern repeated (1)

Example:

<table><tr><th>Colour</th><th>Binary pattern</th></tr><tr><td>Green</td><td>000</td></tr><tr><td>Black</td><td>001</td></tr><tr><td>White</td><td>010</td></tr><tr><td>Red</td><td>011</td></tr><tr><td>Blue</td><td>100</td></tr></table></td><td></td><td>2</td></tr>
<tr><td>2(d)(ii)</td><td>Award one mark for each of:

• 3579 x 6128 x 32 (1)
• ÷ 8 (1)
• + 732 (1)
• ÷ (1000 x 1000) (1)

Examples

$\frac{\frac{3579 \times 6128 \times 32}{8} + 732}{1000 \times 1000}$

$\frac{((3579 \times 6128 \times 32) \div 8) + 732}{1000 \times 1000}$

$\frac{(3579 \times 6128 \times 4) + 732}{1000 \times 1000}$</td><td>• Units are not required
• Equivalent expressions are accepted
• Calculations not explicit but expressed gain the mark
• Award 1 mark for correct calculated answer of 87.73 if no other marks awarded</td><td>4</td></tr>
<tr><td></td><td></td><td>Total for question 2</td><td>15</td></tr>
</table>

Type MS · Expressions for file size and transfer time

Answer 1 of 9 · Question 2 part d.ii · 4 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 2 of 9 · Question 4 part b · 4 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

4(a)	Any **two** from: Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1) Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1) Kilobyte is equivalent to 10^3 bytes, whereas a kibibyte is equivalent to 2^{10} bytes (1) Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1)	Ignore units	2

Type MS · Expressions for file size and transfer time

Answer 2 of 9 · Question 4 part b · 4 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **4 marks** for 19.17 or **One** mark for each of: • Calculates correct bits/bytes(1) • Adds metadata (1) • Calculates correct megabytes (1) • Completely correct response (1) Examples: $\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$ $\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$ ((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB ((2322 x 4128 x 2 bytes per colour) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB	• Arithmetic precedence rules must be clearly followed • Units are not required • Equivalent expressions are accepted • Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark	4

Type MS · Expressions for file size and transfer time

Answer 2 of 9 · Question 4 part b · 4 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Any **two** from: • To save disc/storage space (1) • Reduce transmission time (1) • Meet email attachment size restrictions (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Any **two** from: • Compressing / decompressing the file takes execution time (1) • Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1) • (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1) • Can affect the quality of the file (1) • File size can increase (1)		2

Type MS · Expressions for file size and transfer time

Answer 2 of 9 · Question 4 part b · 4 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award 4 marks for 11 Award 3 marks for a value between 10 and 11 **One** mark for each: • 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1) • + 40 or + 80 (1) • ÷ 1024 (1) • Ceiling or RoundUp (1)	• Arithmetic precedence rules must be clearly followed • Units are not required • Alternative notations for ceiling are accepted • If 16 bits per Unicode character is not awarded, then allow follow through error • Allow follow through error, if interim calculations attempted • Equivalent expressions are accepted	4

Type MS · Expressions for file size and transfer time

Answer 2 of 9 · Question 4 part b · 4 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question Number	Answer	Mark

Type MS · Expressions for file size and transfer time

Answer 3 of 9 · Question 4 part d · 4 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

4(a)	Any **two** from: Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1) Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1) Kilobyte is equivalent to 10^3 bytes, whereas a kibibyte is equivalent to 2^{10} bytes (1) Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1)	Ignore units	2

Type MS · Expressions for file size and transfer time

Answer 3 of 9 · Question 4 part d · 4 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **4 marks** for 19.17 or **One** mark for each of: • Calculates correct bits/bytes(1) • Adds metadata (1) • Calculates correct megabytes (1) • Completely correct response (1) Examples: $\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$ $\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$ ((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB ((2322 x 4128 x 2 bytes per colour) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB	• Arithmetic precedence rules must be clearly followed • Units are not required • Equivalent expressions are accepted • Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark	4

Type MS · Expressions for file size and transfer time

Answer 3 of 9 · Question 4 part d · 4 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Any **two** from: • To save disc/storage space (1) • Reduce transmission time (1) • Meet email attachment size restrictions (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Any **two** from: • Compressing / decompressing the file takes execution time (1) • Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1) • (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1) • Can affect the quality of the file (1) • File size can increase (1)		2

Type MS · Expressions for file size and transfer time

Answer 3 of 9 · Question 4 part d · 4 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award 4 marks for 11 Award 3 marks for a value between 10 and 11 **One** mark for each: • 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1) • + 40 or + 80 (1) • ÷ 1024 (1) • Ceiling or RoundUp (1)	• Arithmetic precedence rules must be clearly followed • Units are not required • Alternative notations for ceiling are accepted • If 16 bits per Unicode character is not awarded, then allow follow through error • Allow follow through error, if interim calculations attempted • Equivalent expressions are accepted	4

Type MS · Expressions for file size and transfer time

Answer 3 of 9 · Question 4 part d · 4 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part d in this mark scheme extract

Question Number	Answer	Mark

Type MS · Expressions for file size and transfer time

Answer 4 of 9 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

12

Type MS · Expressions for file size and transfer time

Answer 4 of 9 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

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Type MS · Expressions for file size and transfer time

Answer 4 of 9 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 / 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Expressions for file size and transfer time

Answer 4 of 9 · Question 4 part b · 4 marks

2020 NOV · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 5 of 9 · Question 1 part g.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part g.ii in this mark scheme extract

Number			
1(a)	**The only correct answer is B** ***A*** *is not correct because computers only understand binary* ***C*** *is not correct because Hex values will be held as binary so take up the same space* ***D*** *is not correct because Hex values are held as binary and so take the same time to execute*		1

Question Number	Answer	Additional Guidance	Mark
1(b)	Award **one** mark for each of: • 5 • A	Must be in the correct order	2

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is B** ***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits* ***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits* ***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits*		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each nibble 0110 0101		2

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark from: • 01000 (1) • 1000 (1)		1

Type MS · Expressions for file size and transfer time

Answer 5 of 9 · Question 1 part g.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part g.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(f)	7		1

Question Number	Answer	Additional Guidance	Mark
1(g)(i)	Award **one** mark from: • The smallest block (of colour) in an image (1) • A single dot in a picture (1) • A picture element (1) • A single element of a digital screen (1)		1
1(g)(ii)	Award **one** mark from: • 8 x 9 (1) • 9 x 8 (1)	Accept 72	1
1(g)(iii)	4		1

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 6 of 9 · Question 2 part g · 4 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part g in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Expressions for file size and transfer time

Answer 6 of 9 · Question 2 part g · 4 marks

2021 JUN · Paper 01 · extract 2 of 5 · find part g in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · Expressions for file size and transfer time

Answer 6 of 9 · Question 2 part g · 4 marks

2021 JUN · Paper 01 · extract 3 of 5 · find part g in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Expressions for file size and transfer time

Answer 6 of 9 · Question 2 part g · 4 marks

2021 JUN · Paper 01 · extract 4 of 5 · find part g in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Expressions for file size and transfer time

Answer 6 of 9 · Question 2 part g · 4 marks

2021 JUN · Paper 01 · extract 5 of 5 · find part g in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 7 of 9 · Question 1 part b.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Expressions for file size and transfer time

Answer 7 of 9 · Question 1 part b.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form 400 x 200 x 12 8 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Expressions for file size and transfer time

Answer 7 of 9 · Question 1 part b.i · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Expressions for file size and transfer time

Answer 7 of 9 · Question 1 part b.i · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 8 of 9 · Question 3 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · Expressions for file size and transfer time

Answer 8 of 9 · Question 3 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

Type MS · Expressions for file size and transfer time

Answer 8 of 9 · Question 3 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · Expressions for file size and transfer time

Answer 8 of 9 · Question 3 part a.ii · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Expressions for file size and transfer time

Answer 9 of 9 · Question 3 part b.ii · 4 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

3(a)(i)	Award **one** mark for each correct hexadecimal character in the correct order: • B9		2
Question	Answer	Additional Guidance	Mark
3(a)(ii)	**The only correct answer is B** *A is not correct because both are processed at the same speed by a computer* *C is not correct because both use the same amount of storage* *D is not correct because hexadecimal uses number base 16*		1
Question	Answer	Additional Guidance	Mark
3(a)(iii)	Award **two** marks for a linked description such as: • (logical/arithmetic) right shift (1) by 2 places (1)		2
Question	Answer	Additional Guidance	Mark
3(b)(i)	Award **one** mark for: Pixel/picture element (1)	Accept additional words. Accept incorrect spelling if word is recognisable as pixel	1

Type MS · Expressions for file size and transfer time

Answer 9 of 9 · Question 3 part b.ii · 4 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
3(b)(ii)	Award **one** mark for each of: • 16x16 pixels (1) • Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1) • Dividing by 8 in the denominator to convert to bytes (1) • Dividing 1000 by the number of bytes used for one image (1) Example: $1000 / \frac{16 \times 16 \times 2}{8}$ bytes Allow follow-through	4 marks for any of: 1000 X 8 / 16 x 16 x 2 8000 / 256 x 2 1000 / 16 x 2 x 2 1000 / 16 x 4 8000 / 512	4
Question	Answer	Additional Guidance	Mark
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1) • requires less storage space (1) encoding 16W for line 1 (1)	Allow valid RLE examples from the image. Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
3(c)	Award **one** mark for each correct cell: **Encrypted text** / **Plain text** RED COT	Ignore capitalisation Must be correct spelling	2
		Total for question 3	14

Type MS · Expressions for file size and transfer time

Answer 9 of 9 · Question 3 part b.ii · 4 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark